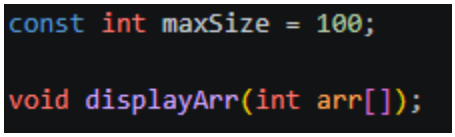
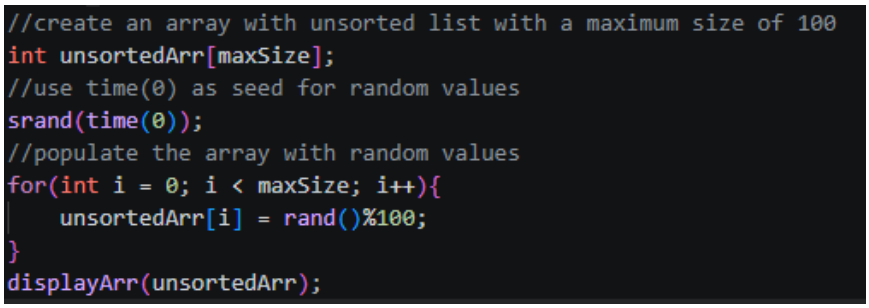
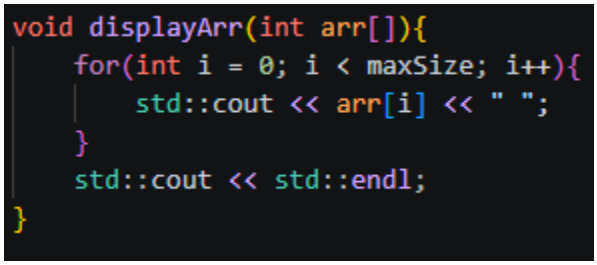
COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 / Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	09/08/2026	**DATE SUBMITTED:**	09/17/2026
NAME:	Lance Cristopher S. Salcedo	**INSTRUCTOR:**	Engr. Jimlord Quejardo

HANDS ON ACTIVITY NO. 7.1 **Sorting Algorithms Pt. 1**

1. PROCEDURE OUTPUT

Code + Console Screenshot	  
Observations	I noticed that it uses an unsorted array with the maximum of 100. It was generated through seed and used for a loop to populate it with random values. The void function also allow us to display it based on the maxSize which is 100.

```
const int maxSize = 100;

void displayArr(int arr[]);
```

```
//create an array with unsorted list with a maximum size of 100
int unsortedArr[maxSize];
//use time(0) as seed for random values
srand(time(0));
//populate the array with random values
for(int i = 0; i < maxSize; i++){
    unsortedArr[i] = rand()%100;
}
displayArr(unsortedArr);
```

```
void displayArr(int arr[]){
    for(int i = 0; i < maxSize; i++){
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
}
```

Table 7-1. Array of Values for Sorting Algorithm Testing

Code + Console Screenshot	Code:

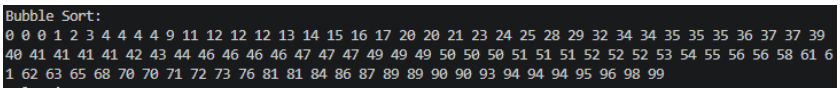
	 Output: 
Observations	I noticed on both of the for loops that it switches i and j. Since bubble sort compares both i and j if swap is needed to implement.

Table 7-2. Bubble Sort Technique

Code + Console Screenshot	Selection Sort:

```
template <typename T>
int routine_smallest(T arr[], int k, int arrSize){
    int position;

    T smallestElement = arr[k];

    for(int j = k + 1; j < arrSize; j++){
        if(arr[j] < smallestElement){
            smallestElement = arr[j];
            position = j;
        }
    }
    return position;
}

template <typename T>
void selectionSort(T arr[], int arrSize){
    int min, temp;

    for(int i = 0; i < arrSize; i++){
        min = routine_smallest(arr, i, arrSize);
        temp = arr[i];
        arr[i] = arr[min];
        arr[min] = temp;
        //std::swap(arr[i], arr[min]);
    }
}
```

Output:

```
Selection Sort:
99 0 0 0 1 2 3 4 4 4 4 9 11 12 12 12 13 14 15 16 17 20 20 21 23 24 25 28 29 32 34 34 35 35 35 36 37 37
39 40 41 41 41 41 42 43 44 46 46 46 46 47 47 47 49 49 49 50 50 50 51 51 51 52 52 52 53 54 55 56 56 58 6
1 61 62 63 65 68 70 70 71 72 73 76 81 81 84 86 87 89 89 90 90 93 94 94 94 95 96 98
```

Observations	I noticed that it uses the smallest_element to make the selection sort work. It also adds a temporary variable which is why 99 was first in the selection sort.

Table 7-3. Selection Sort Technique

Code + Console Screenshot	Insertion Sort:

```
template <typename T>
void insertionSort(T arr[], int arrSize){
    int k = 1, j;
    T temp;

    while(k < arrSize){
        temp = arr[k];
        j = k - 1;
        while(j >= 0 && temp <= arr[j]){
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = temp;
        k++;
    }
};
```

Output:

```
Insertion Sort:
0 0 0 1 2 3 4 4 4 4 9 11 12 12 12 13 14 15 16 17 20 20 21 23 24 25 28 29 32 34 34 35 35 35 36 37 37 39
40 41 41 41 41 42 43 44 46 46 46 46 47 47 47 49 49 49 50 50 50 51 51 51 52 52 52 53 54 55 56 56 58 61 6
1 62 63 65 68 70 70 71 72 73 76 81 81 84 86 87 89 89 90 90 93 94 94 94 95 96 98 99
```

Observations	I observed that insertion sort uses a temporary variable and uses a while loop. In the array it uses the first index of the array which is 0, which is why 0 is in the first in the insertion sort then it make a temporary variable and check whether it is needed to swap if its larger than it.

Table 7-4. Insertion Sort Technique

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Problem:

Candidate 1	Bo Dalton Capistrano
Candidate 2	Cornelius Raymon Agustín
Candidate 3	Deja Jayla Bañaga
Candidate 4	Lalla Brielle Yabut
Candidate 5	Franklin Relano Castro

List of Candidates

Problem: Generate an array A[0…100] of unsorted elements, wherein the values in the array are indicative of a vote to a candidate. This means that the values in your array must only range from 1 to 5. Using sorting and searching techniques, develop an algorithm that will count the votes and indicate the winning candidate.

```cpp
#include <iostream>
#include "supplementaryAlgo.h"

const int maxSize = 100;
void displayArr(int arr[]);

int main(){
    int votes[maxSize];
    srand(time(0));
    for(int i = 0; i < maxSize; i++){
        votes[i] = (rand() % 5) + 1; //generates votes from 1 to 5
    }
    displayArr(votes);

    bubbleSort(votes, maxSize);

    std::string candidates[5]{ //this will input the name of the candidates
        "Bo Dalton Capistrano",
        "Cornelius Raymon Agustin",
        "Deja Jayla Banaga",
        "Lalla Brielle Yabut",
        "Franklin Relano Castro"
    };

    //counts votes
    int counts[5] = {0};
    for (int i = 0; i < maxSize; i++){
        if (votes[i] >= 1 && votes[i] <= 5){
            counts[votes[i] - 1]++; //incrementing the votes
        }
    }

    int maxVotes = counts[0];
    int winner = 0;

    std::cout << "Results of the Elections: \n";
    for (int i = 0; i < 5; i++){
        std::cout << "Candidate " << (i + 1) << ": " <<  candidates[i] <<
        " has " << counts[i] << " votes \n";
        if (counts[i] > maxVotes){ //determines the winner
            maxVotes = counts[i];
            winner = i;
        }
    }

    std::cout << "Winner: " << candidates[winner];

    return 0;
}

void displayArr(int arr[]){
    for(int i = 0; i < maxSize; i++){
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
}
```

Table 7-5. Code of the Algorithm

```
3 4 2 3 3 5 3 3 1 3 2 4 4 4 4 2 5 3 2 4 1 2 3 2 1 2 3 1 2 5 2 3 1 4 2 5 1 1 5 4 4 4 4 4 4 4 5 5 3 3 5 2 2 2 2 3 4 3 4 5
 1 5 4 3 1 4 5 1 1 1 3 3 1 4 5 3 1 5 4 1 4 2 1 5 2 2 4 3 4 1 4 5 4 4 1 2 1 1 3 2
Results of the Elections:
Candidate 1: Bo Dalton Capistrano has 20 votes
Candidate 2: Cornelius Raymon Agustin has 19 votes
Candidate 3: Deja Jayla Banaga has 20 votes
Candidate 4: Lalla Brielle Yabut has 26 votes
Candidate 5: Franklin Relano Castro has 15 votes
Winner: Lalla Brielle Yabut
```

```
4 4 2 5 4 4 5 3 3 5 4 4 2 1 4 2 1 1 2 5 1 3 1 3 3 3 2 4 1 3 5 5 4 4 5 3 2 3 1 4 2 2 5 2 5 3 4 5 2 5 5 5 5 3 1 1 4 3 2 3
 4 4 1 5 2 2 5 3 1 5 5 1 4 3 1 3 5 2 4 5 3 1 5 2 4 3 4 2 5 5 4 2 2 2 5 4 2 4 5 3
Results of the Elections:
Candidate 1: Bo Dalton Capistrano has 14 votes
Candidate 2: Cornelius Raymon Agustin has 20 votes
Candidate 3: Deja Jayla Banaga has 19 votes
Candidate 4: Lalla Brielle Yabut has 22 votes
Candidate 5: Franklin Relano Castro has 25 votes
Winner: Franklin Relano Castro
```

```
3 1 5 5 4 5 5 3 3 3 3 4 2 4 5 2 5 5 5 3 2 1 2 4 4 1 1 5 3 4 3 2 3 5 3 2 3 5 1 1 1 3 4 2 4 3 5 2 3 3 5 3 3 5 3 2 4 3 5 1
 2 2 4 5 5 1 3 2 5 3 3 1 5 1 4 1 2 5 2 5 1 1 3 4 2 1 5 3 1 3 5 5 3 4 5 4 4 1 4 5
Results of the Elections:
Candidate 1: Bo Dalton Capistrano has 17 votes
Candidate 2: Cornelius Raymon Agustin has 15 votes
Candidate 3: Deja Jayla Banaga has 26 votes
Candidate 4: Lalla Brielle Yabut has 16 votes
Candidate 5: Franklin Relano Castro has 26 votes
Winner: Deja Jayla Banaga
```

Table 7-6. Output of the Algorithm

Question:

Question: Was your developed vote counting algorithm effective? Why or why not?

Answer:

Yes, since the bubble sort organizes the mixed array best on the number of candidates although I didn't make it seen. After the bubble sort was able to organize it, the code counts how many are the numbers 1 to 5, then it determines the winner.

Table 7-8. Question & Answer

Conclusion:

I was able to finish the Hands On Activity. I immediately understand how the three types of sorting algorithm works. The one that I used for the voting algorithm which is bubble sort seems effective though I might end up using another sorting algorithm just in case.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **To explain about the observation of certain codes of sorting algorithms.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used Gemini to explain observations and all codes and sentences are mine.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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